FOCUSED POWER BI ENTERPRISE PLAYBOOK

Maximum Impact in Minimum Time

SESSION OVERVIEW

Duration: 60-90 minutes

Scope: All data sources, streamlined execution

Focus: Data modeling → DAX → RLS → Service deployment

What You'll Demonstrate:

- ✅ Complete star schema (all 8 tables)
- ✅ 2 polished report pages (Executive + Detail)
- ✅ 15 essential DAX measures (simple → intermediate)
- ✅ Static RLS (quick implementation)
- ✅ End-to-end Power BI Service setup

Time Allocation:

- Power Query: 10 min (load only, minimal transformation)
- Data Modeling: 15 min (relationships + setup)
- DAX: 15 min (12 core measures)
- Report Pages: 20 min (2 pages)
- RLS: 10 min (static pattern)
- Power BI Service: 15 min (publish + settings)

STAR SCHEMA (ALL TABLES)

Fact Tables

FactSales

100,010 rows

- Source: fact_sales.csv
- Keys: Date, ProductID, CustomerID, RegionID, ChannelID
- Measures: Quantity, UnitPrice, Discount, Cost, Revenue

FactMarketingSpend

30,000 rows

- Source: 6 marketing CSVs appended
- Keys: Date, ProductID
- Attributes: Campaign
- Measures: Spend

FactFinancePlanning

9,600 rows

- Source: finance_planning.xlsx
- Keys: ProductID, MonthKey
- Measures: BudgetRevenue, ForecastRevenue
- Transform: Create MonthKey from Month column

Dimension Tables

DimDate

1,461 rows - Already loaded

Date, Year, Quarter, Month, MonthName

DimProduct

205 rows

ProductID, ProductName, Category, Subcategory

DimCustomer

2,000 rows

CustomerID, CustomerName, Segment

DimRegion

8 rows

RegionID, RegionName, State, Country

DimChannel

4 rows

ChannelID, ChannelName

DimMonth

48 rows - Create in Power BI

MonthKey, MonthStartDate, Year, Quarter, Month, MonthName

Relationships (11 total)

```
FactSales (5 relationships)
 ├──→ DimDate[Date]
 ├──→ DimProduct[ProductID]
 ├──→ DimCustomer[CustomerID]
 ├──→ DimRegion[RegionID]
 └──→ DimChannel[ChannelID]

FactMarketingSpend (2 relationships)
 ├──→ DimDate[Date]
 └──→ DimProduct[ProductID]

FactFinancePlanning (2 relationships)
 ├──→ DimMonth[MonthKey]
 └──→ DimProduct[ProductID]

DimMonth (1 relationship)
 └──→ DimDate[Date] via MonthStartDate (INACTIVE)
```

PHASE 1: POWER QUERY (10 minutes)

Step 1: Load Dimensions (4 tables)

DimDate already loaded - skip

Actions:

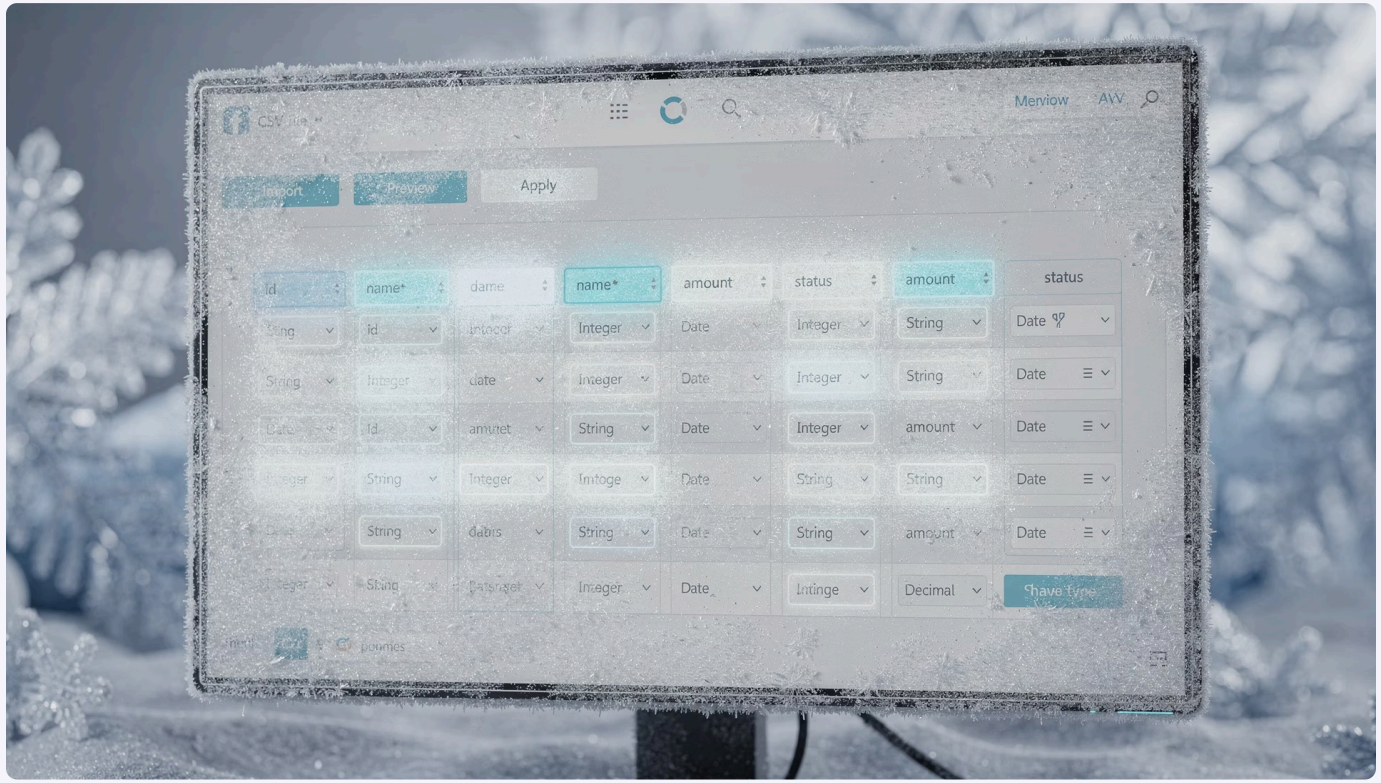
1. Get Data → Text/CSV → dim_product.csv → Load → Rename "DimProduct"
2. Get Data → Text/CSV → dim_customer.csv → Load → Rename "DimCustomer"
3. Get Data → Text/CSV → dim_region.csv → Load → Rename "DimRegion"
4. Get Data → Text/CSV → dim_channel.csv → Load → Rename "DimChannel"

Time: 3 minutes

Step 2: Load FactSales

Actions:

1. Get Data → Text/CSV → fact_sales.csv
2. Rename to "FactSales"
3. Set types: Date (Date), IDs (Whole Number), Values (Decimal)
4. Close & Apply



Time: 5 minutes total

Step 3: Append Marketing Files (Quick Method)

Actions:

1. Get Data → Text/CSV → marketing_spring_sale_2022.csv
2. Transform → Append Queries → Three or more tables
3. Select all 6 marketing files (spring_sale_2022/2023, eofy_blast_2022/2023/2024/2025)
4. Rename to "FactMarketingSpend"
5. Set types: Campaign (Text), Date (Date), ProductID (Whole), Spend (Decimal)
6. Close & Apply

Time: 8 minutes total

Step 4: Load & Transform Finance Planning

Actions:

1. Get Data → Excel → finance_planning.xlsx → Select sheet
2. Rename to "FactFinancePlanning"
3. Add Custom Column:

```
MonthKey = Number.From(Text.Replace([Month], "-", ""))
```

1. Set MonthKey to Whole Number
2. Close & Apply

Time: 10 minutes total (PHASE 1 COMPLETE)

Step 5: Create DimMonth (While in Power Query)

Actions:

1. New Source → Blank Query → Advanced Editor
2. Paste this code:

```
let
    StartDate = #date(2022,1,1),
    MonthList = List.Dates(StartDate, 48, #duration(30,0,0,0)),
    MonthStarts = List.Transform(MonthList, each Date.StartOfMonth(_)),
    UniqueMonths = List.Distinct(MonthStarts),
    ToTable = Table.FromList(UniqueMonths, Splitter.SplitByNothing(), {"MonthStartDate"}),
    SetType = Table.TransformColumnTypes(ToTable, {{"MonthStartDate", type date}}),
    AddMonthKey = Table.AddColumn(SetType, "MonthKey", each Date.Year([MonthStartDate])*100 + Date.Month([MonthStartDate]),
Int64.Type),
    AddYear = Table.AddColumn(AddMonthKey, "Year", each Date.Year([MonthStartDate]), Int64.Type),
    AddMonth = Table.AddColumn(AddYear, "Month", each Date.Month([MonthStartDate]), Int64.Type),
    AddMonthName = Table.AddColumn(AddMonth, "MonthName", each Date.ToText([MonthStartDate], "MMM"), type text)
in
    AddMonthName
```

1. Rename to "DimMonth"
2. Close & Apply

Optional: Skip DimMonth if time is very tight - planning analysis can be excluded from demo

PHASE 2: DATA MODELING (15 minutes)

Step 6: Create All Relationships

What: Build complete star schema

Validate: All 1:*, single direction

Actions (Model View):

FactSales relationships (5):

1. FactSales[Date] → DimDate[Date]
2. FactSales[ProductID] → DimProduct[ProductID]
3. FactSales[CustomerID] → DimCustomer[CustomerID]
4. FactSales[RegionID] → DimRegion[RegionID]
5. FactSales[ChannelID] → DimChannel[ChannelID]

FactMarketingSpend relationships (2):

1. FactMarketingSpend[Date] → DimDate[Date]
2. FactMarketingSpend[ProductID] → DimProduct[ProductID]

FactFinancePlanning relationships (2) - *Optional if skipping DimMonth*:

1. FactFinancePlanning[MonthKey] → DimMonth[MonthKey]
2. FactFinancePlanning[ProductID] → DimProduct[ProductID]

Bridge relationship (1) - *Optional*:

1. DimMonth[MonthStartDate] → DimDate[Date] - **Set to INACTIVE**

Verify each: Many-to-one (1:*), Single direction, Active (except #10)

> **Speaker Note**: "Star schema complete. Facts in center, dimensions around them. All filters flow from dimensions to facts. Single direction prevents circular logic."

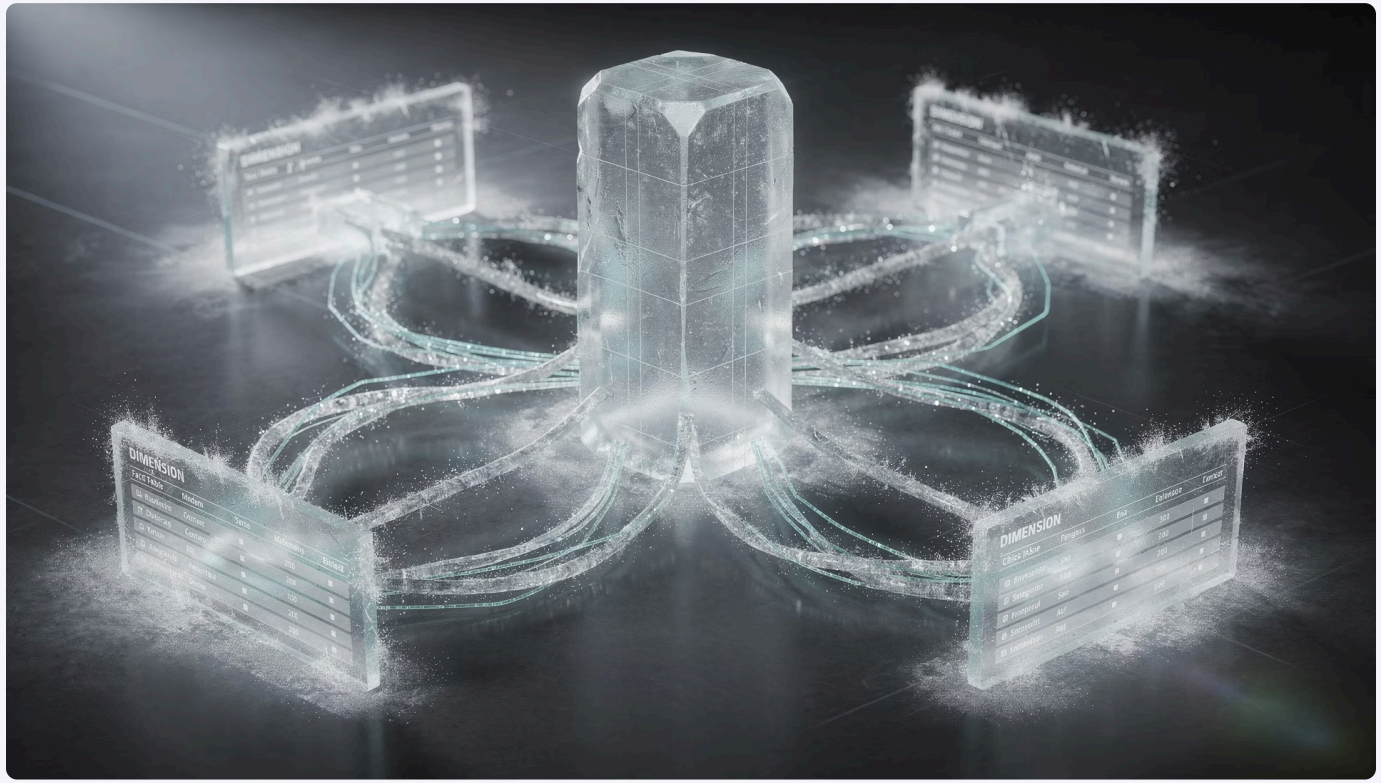
Time: 10 minutes

Step 7: Mark Date Table & Hide Keys

Actions:

1. Right-click DimDate → Mark as date table → Select Date column
2. Select FactSales → Multi-select all ID columns and Date → Hide in report view
3. Select FactMarketingSpend → Hide Date, ProductID
4. Select FactFinancePlanning → Hide ProductID, MonthKey (if using)

> **Speaker Note**: "Marking date table enables time intelligence. Hiding keys forces users to filter via dimension attributes."



Time: 15 minutes (PHASE 2 COMPLETE)

PHASE 3: DAX MEASURES (15 minutes)

Step 8: Create Measures Table

Actions:

1. Home → Enter Data → Blank → Load
2. Rename to "_Measures"
3. Hide Column1

Time: 1 minute

Step 9: Core Business Metrics (8 measures)

Copy-paste these into _Measures table:

Revenue & Profitability:

```
Total Revenue = SUM(FactSales[Revenue])

Total Cost = SUM(FactSales[Cost])

Gross Profit = [Total Revenue] - [Total Cost]

Gross Margin % = DIVIDE([Gross Profit], [Total Revenue], 0)
```

Volume:

```
Total Quantity = SUM(FactSales[Quantity])

Total Transactions = COUNTROWS(FactSales)
```

Average Metrics:

```
Avg Transaction Value = DIVIDE([Total Revenue], [Total Transactions], 0)

Avg Unit Price = DIVIDE([Total Revenue], [Total Quantity], 0)
```

> **Speaker Note**: "DIVIDE with 0 as third parameter handles division by zero gracefully. Production pattern."

Time: 5 minutes

Step 10: Time Intelligence (4 measures)

```
Revenue PY =
CALCULATE(
    [Total Revenue],
    SAMEPERIODLASTYEAR(DimDate[Date])
)

Revenue YoY % =
DIVIDE(
    [Total Revenue] - [Revenue PY],
    [Revenue PY],
    0
)

Revenue YTD = TOTALYTD([Total Revenue], DimDate[Date])

Gross Profit YTD = TOTALYTD([Gross Profit], DimDate[Date])
```

> **Speaker Note**: "SAMEPERIODLASTYEAR only works because we marked DimDate. TOTALYTD accumulates from Jan 1 to current date."

Time: 10 minutes

Step 11: Marketing Metrics (3 measures)

```
Total Marketing Spend = SUM(FactMarketingSpend[Spend])

Marketing ROI = DIVIDE([Total Revenue], [Total Marketing Spend], 0) - 1

ROAS = DIVIDE([Total Revenue], [Total Marketing Spend], 0)
```

> **Speaker Note**: "Cross-table calculation. Revenue from sales, spend from marketing. Product filter aligns them."

Time: 15 minutes (PHASE 3 COMPLETE)

PHASE 4: REPORT DESIGN (20 minutes)

Page 1: Executive Dashboard

Layout (6 visuals, clean and professional):

Row 1 - KPI Cards (4 cards):

Total Revenue

- Format: $#,##0
- Font size: 32pt
- Background: White

Gross Profit

- Format: $#,##0
- Font size: 32pt
- Background: White

Gross Margin %

- Format: 0.0%
- Font size: 32pt
- Background: White

Total Quantity

- Format: #,##0
- Font size: 32pt
- Background: White

Row 2 - Trend Analysis:

5. **Line Chart** - "Revenue Performance"

- X-axis: DimDate[Date] (Year → Month hierarchy)
- Values: Total Revenue, Revenue PY
- Legend: Bottom
- Line thickness: 3
- Data labels: On (current year only)
- Title: "Revenue Trend vs Prior Year"

Row 3 - Category Mix:

6. **Clustered Bar Chart** - "Revenue by Category"

- Y-axis: DimProduct[Category]
- X-axis: Total Revenue
- Data labels: On, inside end
- Sort: Descending by Total Revenue
- Colors: Default theme

Row 4 - Filters:

7. **Slicer Panel** (horizontal layout):

- DimDate[Year] - Dropdown style
- DimCustomer[Segment] - List style
- DimRegion[State] - List style

Formatting:

- Page background: #F5F5F5 (light gray)
- Visual backgrounds: White (#FFFFFF)
- Alignment: Use snap to grid
- Spacing: Consistent 10px margins

Time: 12 minutes

Page 2: Sales Detail Analysis

Layout (4 visuals):

Top Section:

1. **Matrix** - Product Performance

- Rows: Category → Subcategory → ProductName
- Values: Total Revenue, Gross Profit, Gross Margin %, Total Quantity
- Conditional formatting on Gross Margin %:
  - Red: < 20%
  - Yellow: 20-40%
  - Green: > 40%
- Enable drill-down
- Totals: On

Middle Left:

2. **Stacked Bar Chart** - Regional Channel Mix

- Y-axis: DimRegion[RegionName]
- X-axis: Total Revenue
- Legend: DimChannel[ChannelName]
- Data labels: On
- Sort: Descending by Total Revenue
- Title: "Revenue by Region & Channel"

Middle Right:

3. **Donut Chart** - Segment Distribution

- Legend: DimCustomer[Segment]
- Values: Total Revenue
- Detail labels: Category name and percentage
- Title: "Revenue by Customer Segment"

Bottom Section:

4. **Table** - Top Performers

- Columns:
  - DimCustomer[CustomerName]
  - Total Revenue
  - Total Transactions
  - Avg Transaction Value
- Top N Filter: Top 10 by Total Revenue
- Sort: Total Revenue descending
- Conditional formatting: Data bars on Total Revenue
- Title: "Top 10 Customers"

Filters (on this page):

- DimDate[Year]
- DimProduct[Category]

Time: 20 minutes (PHASE 4 COMPLETE)

PHASE 5: ROW-LEVEL SECURITY (10 minutes)

Step 12: Create Static RLS Role

What: Hardcoded region filter (fast implementation)

Why: Show security concept without complexity

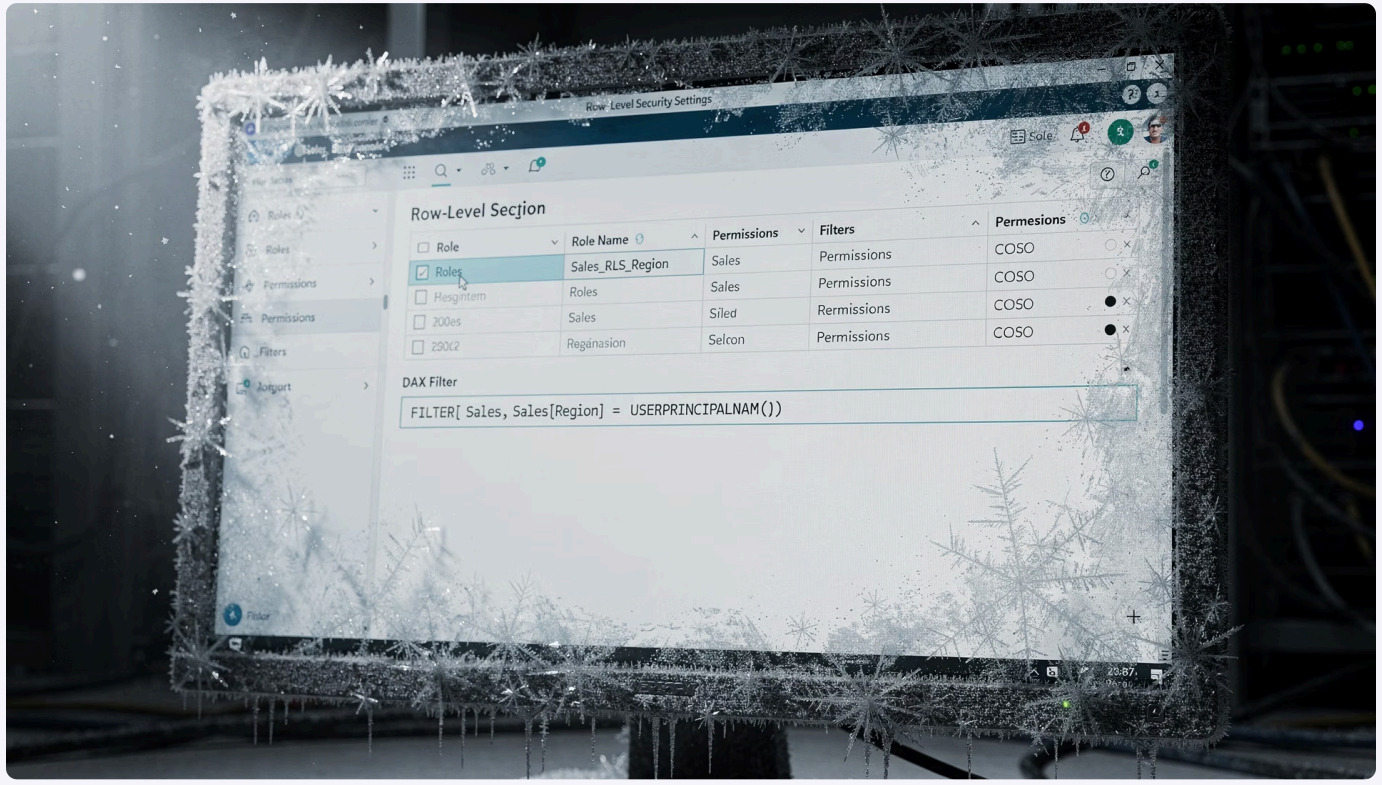
Validate: Test with "View as role"

Actions:

1. Modeling tab → **Manage roles** → **New**
2. Role name: "**NSW Manager**"
3. Select table: **DimRegion**
4. Add DAX filter:

```
[RegionName] = "New South Wales"
```

1. Save



Time: 3 minutes

Step 13: Create Second Role (Optional)

Actions:

1. New role: "**Victoria Manager**"
2. Table: DimRegion
3. Filter:

```
[RegionName] = "Victoria"
```

1. Save

Time: 5 minutes

Step 14: Test RLS

Actions:

1. Modeling → **View as Roles**
2. Check: "NSW Manager"
3. Verify: Only New South Wales data visible
4. Check both pages
5. **Stop viewing**

> **Speaker Note**: "Static RLS is simple but not scalable. In production, use security mapping tables with USERPRINCIPALNAME(). But for demos, this pattern is clear and fast."

Time: 10 minutes (PHASE 5 COMPLETE)

PHASE 6: POWER BI SERVICE (15 minutes)

Step 15: Publish Report

What: Deploy to cloud

Why: Enable sharing and refresh

Validate: Report opens in browser

Actions:

1. Save .pbix file locally first (important!)
2. Home → **Publish**
3. Select workspace:
   - Option 1: My workspace (quick demo)
   - Option 2: Create new workspace (recommended)
4. Wait for "Success" message
5. Click **Open in Power BI**

Time: 2 minutes

Step 16: Configure Dataset Settings

What: Set up data refresh

Why: Keep data current

Validate: Manual refresh succeeds

Actions:

1. Power BI Service → Workspace → **Datasets** tab
2. Click **...** (ellipsis) next to dataset → **Settings**

Data source credentials:

1. Expand "Data source credentials"
2. Click **Edit credentials**
3. Privacy level: Organizational
4. Sign in (OAuth for most sources)

Scheduled refresh:

1. Expand "Scheduled refresh"
2. Toggle: **On**
3. Refresh frequency: **Daily**
4. Time: **6:00 AM** (your timezone)
5. Add owner email for failure notifications
6. **Apply**

Time: 6 minutes

Step 17: Test Manual Refresh

Actions:

1. Datasets tab → Click dataset name
2. Click **Refresh now** icon (top right)
3. Wait for completion (status shows "Last refresh")
4. Check timestamp updated

If refresh fails:

- Check credentials
- Verify source files accessible
- Review error message in refresh history

Time: 8 minutes

Step 18: Assign Users to RLS Roles

What: Apply security in service

Why: Control data access

Validate: Test users see filtered data

Actions:

1. Workspace → **Datasets** tab
2. Click **...** next to dataset → **Security**
3. Select role: **NSW Manager**
4. Under "Members": Add user emails or groups
   - Example: manager.nsw@company.com
5. Select role: **Victoria Manager**
6. Add members: manager.vic@company.com
7. **Save**

> **Speaker Note**: "RLS roles defined in Desktop are enforced in Service automatically. Just assign users to roles."

Time: 11 minutes

Step 19: Share Report

What: Grant viewing access

Why: Users need permissions

Validate: User receives email

Actions:

1. Workspace → **Reports** tab
2. Click report name to open
3. Click **Share** button (top menu)
4. Enter user emails or security groups
5. Permissions:
   - ☑️ Allow recipients to share (optional)
   - ☑️ Send email notification
6. **Grant access**

Important:

- Users inherit RLS roles automatically
- Workspace members see all data (bypass RLS)
- Viewers/consumers get RLS applied

Time: 13 minutes

Step 20: Configure Workspace Settings

What: Set up collaboration

Why: Organize team access

Validate: Permissions align with org

Actions:

1. Workspace → **Access** (top menu)
2. Add users/groups with roles:
   - **Admin**: Full control (IT team)
   - **Member**: Can publish (BI developers)
   - **Contributor**: Can edit in workspace (analysts)
   - **Viewer**: View only (end users)
3. **Add**
4. Close

Time: 15 minutes (PHASE 6 COMPLETE)

VALIDATION CHECKLIST

Power Query ✓

- DimDate: 1,461 rows
- DimProduct: 205 rows
- DimCustomer: 2,000 rows
- DimRegion: 8 rows
- DimChannel: 4 rows
- FactSales: 100,010 rows
- FactMarketingSpend: 30,000 rows
- FactFinancePlanning: 9,600 rows (if included)
- DimMonth: 48 rows (if included)
- No errors in queries

Data Model ✓

- 11 relationships created (10 active, 1 inactive)
- All relationships: 1:*, Single direction
- DimDate marked as date table
- Foreign keys hidden in all fact tables
- No circular dependencies
- Clean star schema layout

DAX ✓

- 15 measures created
- All measures return numeric values
- DIVIDE uses 0 as alternate
- Time intelligence works (test Revenue PY)
- Marketing ROI calculates correctly
- No error messages

Reports ✓

- Page 1: All 7 visuals display data
- Page 2: All 4 visuals display data
- Slicers filter all visuals
- Conditional formatting applied
- Currency formatted correctly ($#,##0)
- Percentages formatted correctly (0.0%)

RLS ✓

- 2 roles created (NSW, Victoria)
- DAX filters applied correctly
- "View as role" tested
- Data filtered to correct regions

Power BI Service ✓

- Report published successfully
- Dataset credentials configured
- Manual refresh succeeds
- Scheduled refresh enabled (daily 6 AM)
- Users assigned to RLS roles
- Report shared with users
- Workspace access configured

COMPLETE DAX LIBRARY (Copy-Paste Ready)

Core Metrics

```
Total Revenue = SUM(FactSales[Revenue])

Total Cost = SUM(FactSales[Cost])

Gross Profit = [Total Revenue] - [Total Cost]

Gross Margin % = DIVIDE([Gross Profit], [Total Revenue], 0)

Total Quantity = SUM(FactSales[Quantity])

Total Transactions = COUNTROWS(FactSales)

Avg Transaction Value = DIVIDE([Total Revenue], [Total Transactions], 0)

Avg Unit Price = DIVIDE([Total Revenue], [Total Quantity], 0)
```

Time Intelligence

```
Revenue PY =
CALCULATE(
    [Total Revenue],
    SAMEPERIODLASTYEAR(DimDate[Date])
)

Revenue YoY % =
DIVIDE(
    [Total Revenue] - [Revenue PY],
    [Revenue PY],
    0
)

Revenue YTD = TOTALYTD([Total Revenue], DimDate[Date])

Gross Profit YTD = TOTALYTD([Gross Profit], DimDate[Date])
```

Marketing

```
Total Marketing Spend = SUM(FactMarketingSpend[Spend])

Marketing ROI = DIVIDE([Total Revenue], [Total Marketing Spend], 0) - 1

ROAS = DIVIDE([Total Revenue], [Total Marketing Spend], 0)
```

SPEAKING NOTES FOR KEY MOMENTS

Model View Demo

Say: "This is a star schema. Facts in the center contain transactions. Dimensions around them contain descriptive attributes. The 1:* symbol means one dimension row relates to many fact rows. Single-direction filtering means filters flow from dimensions to facts, never backward. This prevents circular logic and ensures predictable query performance."

First DAX Measure

Say: "Notice I'm using DIVIDE instead of the forward slash. In production DAX, always use DIVIDE. The third parameter is what shows when the denominator is zero. Without it, you get error messages or blanks. DIVIDE handles it gracefully."

Time Intelligence Demo

Say: "SAMEPERIODLASTYEAR only works because we marked DimDate as a date table. Without that step, this measure would error. Power BI needs to know which table contains your calendar for time intelligence functions to work."

RLS Testing

Say: "We're using static RLS here - the region name is hardcoded. This is fine for demos but not for production. In real implementations, you'd use a security mapping table with USERPRINCIPALNAME() to look up each user's allowed regions dynamically. But for teaching the concept, static is clearer."

Publish to Service

Say: "Always save your .pbix locally before publishing. The cloud version is great for sharing, but your local file is your source of truth. If something goes wrong in the service, you can always republish from Desktop."

Setting Up Refresh

Say: "Data refresh is what keeps your reports current. We're scheduling it for 6 AM before business hours. If refresh fails, the notification email goes to the dataset owner. Common failure causes: source files moved, credentials expired, or gateway offline for on-premises data."

TROUBLESHOOTING QUICK REFERENCE

"Cannot create relationship" error

Cause: Column names don't match or data types differ

Fix: Check exact spelling and set both columns to same type

Time intelligence returns blank

Cause: DimDate not marked as date table

Fix: Right-click DimDate → Mark as date table

Marketing ROI shows wrong values

Cause: Product filter not aligning sales and spend

Fix: Verify both facts have ProductID relationship to DimProduct

RLS not filtering data

Cause: DAX syntax error or wrong column referenced

Fix: Check table[column] syntax, test with "View as role"

Refresh fails in service

Cause: Credentials not configured

Fix: Dataset Settings → Data source credentials → Edit credentials

Users see all data despite RLS

Cause: User has workspace Member/Admin role

Fix: Share report directly (not workspace access) for RLS to apply